

Adding and Subtracting in Scientific Notation

MA.8.NSO.1.5

I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

Last unit you learned to write numbers in scientific notation.

Today you add and subtract them — the powers must match.

Warm-Up

On your own. Four minutes. No notes.

1. Write 0.00062 in scientific notation.
2. How many times larger is 8×10^7 than 2×10^4 ?
3. Find the value: $\sqrt{144}$
4. Find the sum: $0.49 + 1.2$

Warm-Up

Answers.

1. Write 0.00062 in scientific notation. **6.2×10^{-4}**
2. How many times larger is 8×10^7 than 2×10^4 ? **4,000**
3. Find the value: $\sqrt{144}$ **12**
4. Find the sum: $0.49 + 1.2$ **1.69**

Notes

Copy the table.

I. A power of ten is a place value

10^3	10^4	10^5	10^6
thousands	ten thousands	hundred thousands	millions

$$49,500 + 121,000 = 170,500$$

$$4.95 \times 10^4 + 12.1 \times 10^4 = 17.05 \times 10^4 = 1.705 \times 10^5$$

Two numbers can be added only when they carry the same power of ten.

Notes

II. Same power: add or subtract the coefficients

$$(5.2 \times 10^6) + (3.1 \times 10^6) = 8.3 \times 10^6$$

$$(7 \times 10^{-8}) + (6 \times 10^{-8}) = 13 \times 10^{-8} = 1.3 \times 10^{-7}$$

Add or subtract the coefficients. The power of ten does not change.

Then check the coefficient: if it is not between 1 and 10, the answer is not finished.

Notes

III. Different powers: rewrite one of them first

$$(9.4 \times 10^5) - (8.2 \times 10^3)$$

$$= (9.4 \times 10^5) - (0.082 \times 10^5) = 9.318 \times 10^5$$

$$(1.3 \times 10^3) + (3.4 \times 10^5) = 3.413 \times 10^5, \quad \text{never} \quad 4.7 \times 10^8$$

Rewrite one number so both carry the same power of ten, then add or subtract the coefficients.

Exponent up one, decimal point one place left. The value does not move.

Example 1

Same power, and the answer is not finished.

$$(8.6 \times 10^7) + (5.9 \times 10^7)$$

Add. Write the answer in scientific notation.

Example 1

Worked.

$$(8.6 \times 10^7) + (5.9 \times 10^7) = 14.5 \times 10^7 \quad \text{same power: add the coefficients}$$

$$14.5 \times 10^7 = 1.45 \times 10^8 \quad \text{14.5 is not less than 10: point one place left, exponent up one}$$

$$145,000,000 \quad \text{check in standard form}$$

Answer: 1.45×10^8

Your Turn

Same steps, your numbers. Boards up when done.

$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

Your Turn

Answer.

$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

12.3×10^{-3} , then repair

Answer: 1.23×10^{-2}

Example 2

Different powers, and a subtraction.

$$(7.3 \times 10^6) - (4.5 \times 10^5)$$

Subtract. Write the answer in scientific notation.

Example 2

Worked.

$$4.5 \times 10^5 = 0.45 \times 10^6 \quad \text{rewrite over } 10^6$$

$$(7.3 \times 10^6) - (0.45 \times 10^6) = 6.85 \times 10^6 \quad \text{same power: subtract the coefficients}$$

$$7,300,000 - 450,000 = 6,850,000 \quad \text{check in standard form}$$

Answer: 6.85×10^6

Your Turn

Same steps, your numbers. Boards up when done.

$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

Your Turn

Answer.

$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

0.09×10^{-4} comes off 5.6×10^{-4}

Answer: 5.51×10^{-4}

Whiteboards • Question 1 of 9

Boards up on three.

$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

Answer it.

Same power.

Whiteboards • Question 1 of 9

Answer.

$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

add the coefficients

Answer: 7.3×10^5

Whiteboards • Question 2 of 9

Boards up on three.

A stadium sold 4.2×10^4 tickets on Friday
and 3.5×10^4 on Saturday. Each ticket cost 38 dollars.

How many tickets were sold on the two days?

Answer it.

Whiteboards • Question 2 of 9

Answer.

A stadium sold 4.2×10^4 tickets on Friday
and 3.5×10^4 on Saturday. Each ticket cost 38 dollars.

How many tickets were sold on the two days?

$$4.2 + 3.5$$

Answer: 7.7×10^4

Whiteboards • Question 3 of 9

Boards up on three.

$$(6 \times 10^{-5}) + (7 \times 10^{-5})$$

Answer it.

Finish the answer.

Whiteboards • Question 3 of 9

Answer.

$$(6 \times 10^{-5}) + (7 \times 10^{-5})$$

13×10^{-5} is not finished

Answer: 1.3×10^{-4}

Whiteboards • Question 4 of 9

Boards up on three.

$$(5.6 \times 10^8) - (4.9 \times 10^8)$$

Answer it.

Finish the answer.

Whiteboards • Question 4 of 9

Answer.

$$(5.6 \times 10^8) - (4.9 \times 10^8)$$

0.7×10^8 is not finished either

Answer: 7×10^7

Whiteboards • Question 5 of 9

Boards up on three.

$$(1.3 \times 10^3) + (3.4 \times 10^5)$$

Which is the sum?

A. 4.7×10^5

B. 4.7×10^8

C. 3.413×10^5

D. 3.53×10^5

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$(1.3 \times 10^3) + (3.4 \times 10^5)$$

Which is the sum?

A. 4.7×10^5

B. 4.7×10^8

C. 3.413×10^5

D. 3.53×10^5

Answer: C — 3.413×10^5

Whiteboards • Question 6 of 9

Boards up on three.

$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

A. 12.3×10^6

B. 12.3×10^{12}

C. 1.23×10^6

D. 1.23×10^7

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

A. 12.3×10^6

B. 12.3×10^{12}

C. 1.23×10^6

D. 1.23×10^7

Answer: D — 1.23×10^7

Whiteboards • Question 7 of 9

Boards up on three.

$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

Answer it.

Rewrite one of them first.

Whiteboards • Question 7 of 9

Answer.

$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

$$6.4 \times 10^{-3} = 0.64 \times 10^{-2}$$

$$\textbf{Answer: } 7.16 \times 10^{-2}$$

Whiteboards • Question 8 of 9

Boards up on three.

$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of n ?

Answer it.

Rewrite the answer over 10^6 first.

Whiteboards • Question 8 of 9

Answer.

$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of n ?

$$1.02 \times 10^7 = 10.2 \times 10^6$$

Answer: $n = 5.7$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Devon writes $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$.

Name the two mistakes he made, and give the correct sum.

Write your answer in sentences.

Two mistakes, then the sum.

Whiteboards • Question 9 of 9

Answer.

Devon writes $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$.

Name the two mistakes he made, and give the correct sum.

one mistake in the coefficients, one in the exponents

Answer: 3.413×10^5

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Add and subtract numbers written in scientific notation — HUR